1C Order of Operations and Directed Numbers

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Evaluate each expression using the correct order of operations:

(a) $-15 + 6 \times (-4)$ (b) $24 - (-36) \div (-9)$ (c) $-8 \times (-5) + (-28) \div 7$ (d) $50 - 3 \times (-4)^2$

Q2. Evaluate each expression containing brackets and powers:

(a) $(-18 + 6) \div (-3) - 7$ (b) $(-3)^3 + 5 \times (4 - 9)$ (c) $40 \div (3 - 11) - (-2)^3$ (d) $-2 \times [14 - 3 \times (-2)^2]$

Q3. Evaluate each fractional expression by simplifying the numerator and denominator first:

(a) $\dfrac{-24 - 12}{-9}$ (b) $\dfrac{(-5)^2 - 49}{3 - (-5)}$ (c) $\dfrac{-6 \times (-8) - 18}{2 - 17}$ (d) $\dfrac{(-4)^3 + 20}{(-3)^2 + 2}$

Q4. Given $p = -4$, $q = 3$, and $r = -5$, evaluate:

(a) $p^2 - 2qr$ (b) $3p - q^2 + r$ (c) $(p - r)^3 - q^2$ (d) $\dfrac{p^2 + r^2 - 1}{p - q - 1}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. At Mount Hotham, the temperature was -4°C for 3 hours, -2°C for 5 hours, and $+3^\circ\text{C}$ for 2 hours. Write a single expression and calculate the mean temperature across this 10-hour period.

Q6. A Melbourne share trader holds 400 shares in a mining company that drop by $\$3$ per share, and 250 shares in a tech company that rise by $\$6$ per share. She also pays $\$40$ in brokerage fees. Find her net financial gain or loss.

Q7. When converting a temperature of $F = 14^\circ\text{F}$ to degrees Celsius using $C = \dfrac{5(F - 32)}{9}$, evaluate the formula step by step to find C.

Q8. A diver starts at $-12\ \text{m}$, descends at $4\ \text{m/min}$ for 6 minutes, pauses, and then ascends at $3\ \text{m/min}$ for 9 minutes. Evaluate $-12 + 6(-4) + 9(3)$ to find the diver's final position.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Insert one pair of brackets into the expression $-18 \div 3 - 9 + (-2)^3$ so that the expression evaluates to -5. Show your full working.

Q10. Sam evaluates $16 - 4 \times (-3)^2$ as $12 \times 9 = 108$, while Priya evaluates it as $16 - (-12)^2 = 16 - 144 = -128$. Explain the specific order-of-operations error each student made and find the correct value.